

Dark Fibre

Optical Measurements Lead · Pellow Head, twelve days with a cable down

12 DAYS · 40 STOPS · GRADE 12 · DARKFIBRE

THE OPENING CARD

Nineteen days ago the light stopped coming back, and the cable at Pellow Head went dark. Every call on it now runs one spare route, and that route has no spare. You are the optical measurements lead, and the repair ship digs where this station says. It is on hire from the thirteenth, at forty thousand a day, for one crossing. Its grapnel drags a lane five hundred metres wide, and a chart out by more than that cuts the cable.

What stays in the glass

BEFORE THE DAY — TRIAL

Walk the station before the ship sails

Five buildings, a duct trench and a beach manhole, all of it on sand, and the splice trailer is not parked where the drawing puts it.

PLAN CARD 166W

Monday. Twelve working days, and then the ship sails for wherever this station says. Pellow Head is where a submarine cable comes ashore. Nineteen days ago the span out to the first repeater lost four decibels overnight. A decibel is how loss is counted here. Ines Sarraf, the chief transmission engineer, moved the traffic off the cable that night. Gerald Okonkwo, the station chief, laid this cable eleven years ago. He keeps saying that an instrument reports in its own units, not in yours. On the twelfth day you hand the repair chart to Elias Trewin, the cable engineer on the ship. It has to say three things. How far out the fault is. Where that lands along the route rather than along the glass. And what broke. Today you settle why light stays inside a fibre at all. You also work out how much light a termination can get in. Then you hear what the trace in the test room does and does not say.

BRIEFING 15W

Nineteen days with a cable down, and the ship is on charter from the thirteenth.

WHAT YOU DECIDE 18W

Establish why light stays in a fibre at all, and how much of it can be got in.

1.1 Seven degrees of everything

CABLE TERMINATION ROOM · A ROOM · DERIVE

SCENE 40W

Halvard Nilsen, the cable termination technician, has the fibre's two indices on the bench: 1.4682 in the core and 1.4629 in the cladding. He wants the widest cone of light that will stay in the core once it is in.

GUIDE 70W

You build the working a line at a time, choosing the expression the line above gives you. Start at the boundary between the core and the cladding and ask what angle just fails to cross it. Turn that into an angle from the axis rather than from the boundary, then take it out through the end face into air. Two of the three steps are the same law used twice.

ASK 15W

What is the widest angle from the axis that a termination may launch light at?

BEHIND THE BUTTON 166W

Why the two angles are complementary. A ray's angle at the core-cladding boundary is measured from the normal to that boundary. That normal runs across the fibre. Its angle from the axis is measured along the fibre instead. The two add to ninety degrees. Confusing them is the commonest way this derivation goes wrong.

Why the answer is so small. The two indices differ in the third decimal place, and that is deliberate. A small difference makes a narrow cone. A narrow cone keeps every ray on nearly the same path, so the rays arrive nearly together. The alternative is a fibre that accepts far more light and spreads every pulse.

What the number is for. Seven degrees is what a termination has to launch inside. It is also why a fibre is joined by welding rather than by pointing one at another. A misalignment of a fraction of a degree throws light outside the cone. Light outside the cone is gone by the first kilometre.

TAKES AS READ

a ray at a boundary is measured from the normal to that boundary · total internal reflection and the critical angle — taken as read

VERDICT 94W

The cone is about 7.2 degrees in air. Total internal reflection needs a ray to meet the core-cladding boundary beyond the critical angle, measured from the normal. For these indices that critical angle is 85.1 degrees. From the axis, that is 4.9 degrees. Snell's law at the end face converts that internal angle into an external one. The core is denser, so the external cone is wider than the internal one. Numerical aperture is the sine of it, 0.125. It is also the root of the difference of the squares of the two indices.

TAKEAWAY 19W

A fibre's acceptance cone is fixed by its two indices alone, and light launched outside it is lost immediately.

1.2 What came back, and when

TEST ROOM · A PERSON · CHOICE

SCENE 35W

Sarraf has the trace from the night of the loss pinned up. A sloping line, with a step in it nineteen days old. The scale is decibels down the side and kilometres along the bottom.

GUIDE 53W

Four accounts of what that step is. Ask of each whether the instrument could know it, given that all it has is how much light came back and how long it took. The distinction that decides it is between a quantity the instrument measures and a quantity somebody has to convert it into.

ASK 11W

What does the step in the trace establish on its own?

TAKES AS READ

a reflectometer sends a pulse and records what returns against time · time of flight, and turning a delay into a distance — taken as read · power in decibels, and why a loss budget adds — taken as read

VERDICT 82W

It establishes that something 4.1 decibels' worth happened at a particular round-trip delay. Everything else is conversion. The delay becomes a distance only once a speed is assumed. The distance becomes a seabed position only once the cable's slack is allowed for. What the step cannot say at all is what the event is. A break, a bend and a failing joint all put a step in a trace. Telling them apart needs the reflection beside the loss, not the loss alone.

TAKEAWAY 22W

An instrument measures in its own units, and every step from there to a decision is a conversion somebody has to make.

1.3 Eleven joints, eleven years old

SPLICE TRAILER · A ROOM · CHOICE

SCENE 38W

Priya Ramnarine, the chief splicer, has the shore-end record open. Eleven mechanical joints were made on this cable when it was landed. Each holds two ends together in a gel. All eleven are as old as the cable.

GUIDE 61W

Ramnarine's eleven joints each hold two cut ends of glass a hair apart, with gel in the gap. Ask what the gel is touching, and what would be there instead without it. A joint can lose light quietly. It can also send light back the way it came. Only one of those shows up on a trace taken from this building.

ASK 9W

What does the gel in a mechanical joint do?

TAKES AS READ

light is partly reflected wherever the index changes abruptly

VERDICT 86W

It removes the index step. Two glass ends held a hair apart have air between them. Each glass-air boundary reflects a few per cent of the light straight back. A gel of the same index as the glass leaves nothing for the light to notice. A dried joint is therefore both lossy and reflective, and the reflection is what shows on a trace. That is the signature this fortnight turns on. A break reflects, a bend does not, and a dried joint does both a little.

TAKEAWAY 21W

A joint is judged by what it does to the light rather than by how it holds two pieces of glass.

END OF THE DAY 17W

Light stays in a fibre because of two indices, and everything about a launch follows from them.

A delay, and what it is worth

PLAN CARD 139W

Tuesday. The position on the ship's chart is a number somebody copied off an instrument. Sarraf's trace puts the step at a round trip delay of 828.6 microseconds. The software beside it prints a distance in kilometres instead. Okonkwo will not sign a chart when he cannot see the working. Aline Duarte, the receiver engineer, has raised a second worry. The software's index setting has not been checked since the instrument was new. The index is the number that says how slowly light runs in glass. Today you get the distance out of the delay yourself. You also work out how fast light really moves in this glass. Then you settle what light is being counted as at the far end of the fibre. Eleven years of slack cable sit between this number and a place on the seabed.

BRIEFING 17W

The whole fortnight comes down to one number on a chart, and it has never been derived.

WHAT YOU DECIDE 15W

Turn the trace's delay into a distance, and say what speed that distance rests on.

2.1 Eight hundred and twenty-eight microseconds

TEST ROOM · A ROOM · DERIVE

SCENE 31W

The step in the trace sits at a round-trip delay of 828.6 microseconds. The core's index is 1.4682. Sarraf wants the distance derived rather than read off the instrument's own display.

GUIDE 53W

You build the working a line at a time. Light in glass is slower than light in empty space by exactly the index, and the pulse has been out and back rather than out. Both of those are factors, and getting either the wrong way round gives an answer with the right units.

ASK 12W

How far along the fibre is the event that made the step?

BEHIND THE BUTTON 161W

Why the factor of two. A reflectometer measures the light that comes back to it, so every delay it reports is a return journey. Leaving the two out doubles every distance. So the first thing to check on any trace is whether the far end of the fibre appears at its own length or at twice it.

Why the index divides. Light covers less ground per second in glass than in vacuum, so the same delay corresponds to a shorter distance. Multiplying by the index instead gives 182 kilometres, which is longer than the span and would put the fault past the first repeater.

What this distance is and is not. It is a distance along the fibre. The fibre is longer than the cable's route, because it is wound helically inside the cable. The cable is longer than the seabed distance, because it was laid with slack. Both corrections come later in the fortnight, and both make the number smaller.

TAKES AS READ

a reflectometer reports the delay of a pulse that has been out and back

VERDICT 87W

The event is about 84.6 kilometres along the fibre. Light travels at the vacuum speed divided by the index. For this glass that is 2.042×10^8 metres a second. The pulse made the journey twice, so the distance is half the product of speed and delay. Both factors matter and both are easy to invert. Multiplying by the index rather than dividing gives 182 kilometres. Forgetting the return journey gives 169. This is a distance along the fibre, and not yet a position on the seabed.

TAKEAWAY 25W

A time of flight gives a distance along the fibre, and only after the speed in the glass and the return journey are both applied.

2.2 Two hundred thousand kilometres a second

CABLE TERMINATION ROOM · A PERSON · BALLPARK

SCENE 31W

Nilsen has three indices written on the bench: the core, the cladding, and the group index the instrument's software is set to. He wants the speed that goes with the core.

GUIDE 66W

Four quantities are offered and two of them make the answer. The index to use here is the one belonging to the glass the light is in, which is the core. The cladding's index is on the bench because it belongs to the last derivation. The group index is there because it belongs to the next one. The vacuum speed is the only other thing needed.

ASK 11W

How fast does light travel in the core of this fibre?

TAKES AS READ

the index of a material is the factor by which light is slowed in it

VERDICT 83W

Light makes about 2.042×10^8 metres a second in this core. That is 68 per cent of its speed in vacuum. An index is defined as exactly that ratio, so the arithmetic is a single division. The difficulty is never the division. It is which of three indices belongs in it. The core's is the one the light is travelling in. The distinction between it and the group index is worth a kilometre on the ship's chart. It arrives on day ten.

TAKEAWAY 21W

A refractive index is a ratio of speeds, so dividing the vacuum speed by it is the whole of the arithmetic.

2.3 A few hundred of something

RECEIVER LAB · A ROOM · CHOICE

SCENE 34W

Duarte has the receiver's own record on the bench. It gives the power arriving at the end of the span, in billionths of a watt. It also gives the rate the bits arrive at.

GUIDE 41W

All four options describe what a receiver is dealing with. Ask of each whether it is a statement about a wave, about a stream of arrivals, or about the electronics. The answer decides which physics the error rate is limited by.

ASK 12W

What is the receiver working with at the end of the span?

TAKES AS READ

light can be described as a wave and as a stream of arrivals

VERDICT 81W

It is working with a wave of a definite wavelength, arriving as a countable number of pieces. Both halves are needed. The wavelength decides how much energy each arrival carries, and how the fibre treated it on the way. The count decides how steady the arrival rate is. At a few hundred arrivals a bit, the count is what limits the error rate. No amount of better electronics changes it. That is a statement about light rather than about the receiver.

TAKEAWAY 23W

A wave has a wavelength and a frequency; a stream of arrivals has a count, and both descriptions are needed at a receiver.

END OF THE DAY 18W

A delay becomes a distance only through a speed, and the speed is a property of the glass.

What comes straight back

PLAN CARD 152W

Wednesday. The question in the test room has changed from where to what. Ramnarine says a break in the glass throws light straight back at the instrument. A bend does not. So the shape of the trace should say which of the two this is. Sarraf's trace shows a step with a small spike on it. Nobody has measured that spike yet. Tobias Meier, the test room technician, has eleven years of traces taken at two wavelengths. He has said all week that the pair tells you more than either one alone. Today you work out what a break sends back. You also find the wavelength a kilometre of this fibre loses least at. Then you hear from Dr. Yuki Tanaka, the amplifier specialist, what the pump telemetry does and does not report. Whatever the ship is told to look for, it has to be told what kind of thing it is.

BRIEFING 16W

A break reflects and a bend does not, and nobody has put a number on either.

WHAT YOU DECIDE 15W

Work out how much light a break returns, and find the wavelength this fibre likes.

3.1 Three and a half per cent, straight back

SPLICE TRAILER • A ROOM • DERIVE

SCENE 38W

Ramnarine has two cleaved ends on the bench, one dry and one in gel. She wants the fraction of the light each of them sends straight back, in decibels, so that a trace can be read against it.

GUIDE 61W

You build the working a line at a time. What decides the reflection is the size of the step in index at the boundary. Both ends of that step matter. Glass to air is a large step; glass to gel is almost none. Take the fraction first, then put it into decibels, where a small fraction becomes a large negative number.

ASK 14W

What fraction of the light does a dry cleaved end send back, in decibels?

BEHIND THE BUTTON 173W

Why the step is squared. The reflected amplitude goes as the difference over the sum of the two indices. Power goes as the square of amplitude. So a step of one part in eight reflects about one part in seventy of the power. Forgetting the square gives 19 per cent rather than 3.6.

Why the gel matters so much. Its index is within 0.01 of the glass, so the step is a hundredth of the glass-air step. The reflection is then ten thousand times smaller: –51 decibels instead of –14. That is the difference between a joint that shows on a trace and a joint that does not.

What this settles about the fault. A break with a clean face returns about 3.6 per cent. A bend returns nothing at all, because there is no step anywhere — the light simply leaves the core. The trace shows a step with a spike about a fortieth of the size of a clean break's. That is neither, and it is the finding of the fortnight.

TAKES AS READ

power is proportional to the square of a wave's amplitude • interference of two beams, and the path difference that decides it — taken as read • snell's law at a boundary — taken as read

VERDICT 89W

A dry end returns about 3.6 per cent of the light, or –14.4 decibels. The reflected amplitude is the difference of the two indices over their sum, and power is amplitude squared. So a step from 1.468 to 1.000 returns $(0.468/2.468)^2$ of the power. In gel the same expression gives 7.8×10^{-6} . The step is a hundredth the size, and the square makes that ten thousand. The point is not the numbers. It is that the two cases differ by 37 decibels, which any trace can tell apart.

TAKEAWAY 25W

Reflection at a boundary depends on the size of the index step, so the same loss can come back as a mirror or as nothing.

3.2 A window, and the peak beside it

TEST ROOM · A ROOM · SWEEP

SCENE 38W

Meier has the loss set up so that the instrument can be tuned across its whole range while it measures a kilometre of spare fibre on the drum. Nothing has been swept on this station for eleven years.

GUIDE 65W

Move the tuning across the range and read the loss the drum returns at each setting you actually stop at. Only the points you visit are plotted, so the curve is what you have built rather than what the instrument drew. The criterion is the lowest loss per kilometre; note also that there is a narrow rise somewhere in the middle that is not scattering.

ASK 13W

A kilometre of this fibre loses least at one setting. Where is it?

TAKES AS READ

scattering falls steeply as the wavelength gets longer

VERDICT 89W

The loss is lowest near 1555 nanometres, at about 0.19 decibels a kilometre. Two things are happening at once. Scattering by the tiny irregularities of the glass falls steeply as the wavelength grows, as the fourth power of it. That is the whole downward slope. Absorption by the glass itself rises at the long end. Between them is a window. The narrow spike in the middle is water trapped in the glass, absorbing at its own wavelength. That is a property of the molecule, not of the fibre's quality.

TAKEAWAY 16W

A fibre has a window where scattering has fallen away and absorption has not yet begun.

3.3 Current, and not light

AMPLIFIER AND POWER HALL · A PERSON · CHOICE

SCENE 25W

Tanaka has repeater 6's telemetry on the screen: a pump current, a case temperature and a supply voltage, four years of them, all inside limits.

GUIDE 53W

Tanaka has a current, a temperature and a voltage. There is no measurement of light anywhere in the record. Gain depends on the population inversion — how many erbium atoms sit in the upper level, waiting. Ask of each option what would have to be true for a current alone to establish it.

ASK 9W

What does four years of good pump telemetry establish?

TAKES AS READ

an amplifier works by holding more atoms excited than not · atomic energy levels, and the wavelengths they allow — taken as read

VERDICT 90W

It establishes that the pump laser is being driven. Gain depends on the population inversion — on more erbium atoms being in the upper state than the lower. And the inversion depends on the pump's optical output, not on its drive current. A laser diode's output falls with age at a constant current. That is exactly the failure that leaves telemetry looking perfect. Nothing on that screen reaches the number that matters. That is why the last fortnight has been spent inferring it from the far end of a fibre.

TAKEAWAY 20W

A pump current says power is going in; the gain depends on how many atoms that power is actually exciting.

3.4 Two hundred thousand kilometres a second — Review

CABLE TERMINATION ROOM · A ROOM · CALLBACK · BALLPARK

SCENE 34W

Okonkwo, the station chief, has the spare drum's certificate out. It is a different make of fibre, wound eleven years ago as the repair length for this span, and its core index is 1.4571.

GUIDE 64W

Three indices are on the bench and only one belongs in this division. You want the index of the glass the light will actually travel in. That is the repair fibre's core. The span's own core is here because the two get compared once the repair length is in. The cladding is here because it belongs to a boundary rather than to a path.

ASK 12W

How fast does light travel in the core of the repair fibre?

BEHIND THE BUTTON 110W

Why the answer is close to the span's own and still worth taking. The two glasses differ in the third decimal place. That is under one per cent of a speed, and about seven hundred metres over a span of this length. That is inside a grapnel run and outside the tolerance the ship is being given.

What an index actually is. It is the vacuum speed divided by the speed in the material, so it is a ratio of two speeds and carries no units of its own. Anything that comes out of it with a unit attached has had a unit invented for it somewhere in the working.

TAKES AS READ

the index of a material is the factor by which light is slowed in it

VERDICT 89W

Light makes about 2.058×10^8 metres a second in this glass, one per cent faster than in the span it will be joined into. An index is the ratio of the vacuum speed to the speed in the material. The arithmetic is therefore one division, and the whole difficulty is which of three indices goes into it. A repair length of different glass puts a short stretch of the span at a different speed. Every later distance is worked out as though the whole fibre were one glass.

TAKEAWAY 25W

A speed in a material comes out of one division, and the work is deciding which of the indices on the bench belongs in it.

END OF THE DAY 17W

The size of the reflection at an index step is what tells a break from a bend.

What the span can afford

BEFORE THE DAY — TRIAL

The dune track out to the bay

The radiography bay is three hundred metres out in the dunes because of what is in the pot, and the track shifts with the sand. Take it once before you have to, because Nadia Farouk, the radiographer, will be waiting on a film and the exposure cannot be interrupted half way.

PLAN CARD 135W

Thursday. For the first time the argument in the test room has a number in it. Sarraf has said all week that the trace is a measurement. Send the ship to the kilometre it names. Okonkwo says a better number is worth waiting for. The arithmetic decides it today. Whether the span still works depends on what the loss adds up to. That total is set against what the receiver can accept. Nadia Farouk, the radiographer, has the housing records the ship will need either way. Today you total the loss budget. You also settle what attenuation per kilometre means. Then you see what a radiography record has to carry. A week spent hunting a better position is a week the traffic sits on one spare route. That route has no spare of its own.

BRIEFING 13W

Four decibels arrived overnight, and nobody has said what the span had spare.

WHAT YOU DECIDE 15W

Add up the loss along the span and see what is left of the allowance.

4.1 Twenty-one point eight

TEST ROOM · A ROOM · DERIVE

SCENE 35W

The span is 84.6 kilometres of fibre at 0.19 decibels a kilometre, with eleven joints at 0.15 each, and the new event at 4.1. Sarraf wants the total against the receiver's allowance of 24 decibels.

GUIDE 59W

You build the working a line at a time. Every loss here is quoted in decibels, and decibels are logarithms, so the stages add rather than multiply. Take the span first, then the joints, then the event. Only at the end turn the total into a power ratio. That is where the size of what has happened becomes visible.

ASK 13W

What is the total loss on the span, and what margin is left?

BEHIND THE BUTTON 155W

Why a logarithm makes a sum. Each stage multiplies the power by a fraction, and the logarithm of a product is the sum of the logarithms. So a chain of fifteen losses becomes an addition of fifteen numbers, which is the whole reason the unit exists.

What 21.8 decibels is as a ratio. It is a factor of 151: for every hundred and fifty-one photons launched, one arrives. The receiver's allowance of 24 decibels is a factor of 251. What is left is a margin of 2.2 decibels. That is a factor of 1.66, and it is not much.

Why this settles today's argument. With 2.2 decibels of margin the span is working, and it is one small event away from not working. That is an argument for sending the ship on the best number available, rather than for spending a week refining it. That is Sarraf's position, and on today's numbers she is right.

TAKES AS READ

a loss in decibels is ten times the logarithm of the power ratio

VERDICT 91W

The span loses 21.8 decibels and has 2.2 left. Decibels are logarithms of power ratios, so the fifteen stages add. That is 16.1 for the fibre itself, 1.65 for the eleven joints and 4.1 for the new event. Converting the total back gives a factor of 151, against the receiver's allowance of a factor of 251. A margin of 2.2 decibels is a third of one bad joint. That is why the traffic was moved on the night. And it is why a week of refinement is a week of exposure.

TAKEAWAY 21W

A loss budget is an addition because decibels are logarithms, and the margin left is what a span has in hand.

4.2 The same fraction every kilometre

AMPLIFIER AND POWER HALL · A PERSON · CHOICE

SCENE 42W

Bruno Salas, the power feed engineer, has the span's loss written up beside the feed board in the generator house, as a rate. A technician has asked him a question. Do the tenth kilometre and the eightieth take out the same amount?

GUIDE 53W

Four readings of what a loss quoted per kilometre asserts. Ask of each whether it is about an amount of power or a fraction of it. The difference decides whether a long span ends with a little light or with none. It also decides whether adding an amplifier half way is worth anything.

ASK 11W

What does 0.19 decibels a kilometre assert about the eightieth kilometre?

TAKES AS READ

a fixed loss per unit length is a fixed fraction per unit length · power in decibels, and why a loss budget adds — taken as read

VERDICT 78W

It asserts that the eightieth kilometre takes the same fraction as the first, which is about 4.3 per cent, and therefore far less power. That is what a constant figure in decibels means: equal lengths remove equal fractions, so the power falls exponentially. It never reaches zero. So the limit on a span is the receiver's sensitivity, not the fibre running out of light. And an amplifier half way along is worth exactly the loss it makes up.

TAKEAWAY 23W

A rate quoted per unit length is a fraction rather than an amount, and the two behave nothing alike over a long path.

4.3 Four things written on the corner

RADIOGRAPHY BAY • A ROOM • PROTOCOL

SCENE 31W

Farouk has three radiographs of the same weld on the bay bench, taken over two years. All three show the weld. Only one of them has anything written on its corner.

GUIDE 54W

Each thing on the left is written on the corner of a good radiograph. Match it to what somebody comparing two films would use it for. What separates them is which of them is about the source, which about the geometry, and which about the film. Comparing two films means holding all three constant.

ASK 12W

Match each note on a radiograph to what it is there for.

TAKES AS READ

two records can only be compared if the conditions they were taken under are known • power in decibels, and why a loss budget adds — taken as read

VERDICT 79W

Four numbers, four different jobs. The source's age gives its strength without a measurement, because activity falls on a known schedule. The distance fixes the spreading, which goes as the inverse square. The exposure time is the one thing under the radiographer's control and it has to grow as the source decays. And the steel thickness decides how much arrives at all, exponentially. Take any one of them away and two films of the same weld cannot be compared.

TAKEAWAY 20W

A record without its conditions is a picture; with them it is a measurement that can be compared with another.

4.4 A few hundred of something — Review

RECEIVER LAB · A ROOM · CALLBACK · CASEBOOK

SCENE 35W

Karl Osgood, the photodiode technician, has four facts about the far end pinned over the bench, collected on different days by different people. None of them was written down with this fortnight's argument in mind.

GUIDE 57W

Four observations and four accounts of what each one is a statement about. A wave has a wavelength, and everything the glass does to it on the way follows from that. A stream of arrivals has a count, and a count has a scatter of its own that nothing downstream removes. Two of these belong to each.

ASK 14W

Match each fact about the far end to what it is a statement about.

BEHIND THE BUTTON 98W

Why the distinction is worth making at a receiver. It decides what better equipment could change. A loss that belongs to the wavelength is fixed by choosing a different channel or a different glass. A limit that belongs to the count is fixed only by sending more light.

Why both descriptions are needed at once. The wave picture says how the fibre treated the signal on the way and what the amplifier could do to it. The arrival picture says how steady the number reaching the diode is, which is what an error rate is a statement about.

TAKES AS READ

light can be described as a wave and as a stream of arrivals

VERDICT 83W

Two of the four are about the wave and two are about the stream, and the fibre only ever acts on the first pair. A wavelength is what the transmitter is locked to, and how much a kilometre of glass scatters depends steeply on it, so the colour decides the loss. The other two are counting. Arrivals scatter about their average whatever the electronics are doing, and halving the power halves the number counted for each bit, which raises the errors by itself.

TAKEAWAY 16W

Which description an observation belongs to decides whether better equipment at this end could change it.

END OF THE DAY 19W

Losses in decibels add, so a budget is a sum and what is left of it is a margin.

How much is one photon

PLAN CARD 128W

Friday. Duarte has asked a question the test room cannot answer in decibels. Say the power arriving is seven billionths of a watt. How many pieces of light is that a second? And how many of them make up one bit? Tanaka needs the same count from the other end. An amplifier's gain is photons out against photons in. A photon is one piece of light. Nilsen has the launch power at the termination and has never turned it into a count. Today you work out the energy of one photon at the wavelength this cable runs at. You also count the arrivals inside one bit. Then you settle why a pulse comes back wider than it left. Every limit a receiver has is a statement about counting.

BRIEFING 16W

The receiver is working with a few thousand arrivals a bit and nobody has counted them.

WHAT YOU DECIDE 18W

Work out what a piece of light at this wavelength is worth, and how many make a bit.

5.1 Eight tenths of an electron volt

AMPLIFIER AND POWER HALL · A ROOM · DERIVE

SCENE 38W

Tanaka has the two wavelengths this amplifier works at on the board and Planck's constant beside them. She wants the energy of one photon at 1550 nanometres, in electron volts, and how many of them a milliwatt is.

GUIDE 61W

You build the working a line at a time. A photon's energy depends only on its colour, and the two ways of writing that — through the frequency or through the wavelength — are the same statement. Get the energy in joules first, then convert to electron volts, then divide a power by it to turn watts into arrivals a second.

ASK 17W

What is the energy of a photon at 1550 nm, and how many are in a milliwatt?

BEHIND THE BUTTON 171W

Why the wavelength divides. Frequency and wavelength are inversely related through the speed of light, so a longer wavelength is a lower frequency and a smaller energy. A photon at 980 nanometres carries 1.27 electron volts, against 0.80 at 1550. That is the whole reason an amplifier can be pumped at the shorter wavelength and give out at the longer.

Why the electron volt is the natural unit. A joule is a household quantity and a photon is not. A photon's 1.28×10^{-19} joules is hard to hold in the head. Its 0.80 electron volts can be compared straight away with the band gaps of the materials a detector is made of.

What the arrival rate is for. A milliwatt is 7.8×10^{15} photons a second, which sounds like an inexhaustible supply. Divide it by ten billion bits a second, and reduce it by a factor of 151 along the span. Then it is a few thousand a bit, and a few thousand of anything counted has a scatter.

TAKES AS READ

light of one wavelength arrives only in whole quanta of a fixed energy

VERDICT 89W

A photon at 1550 nanometres carries 1.28×10^{-19} joules, or 0.80 electron volts. A milliwatt is 7.8×10^{15} of them a second. The energy comes from Planck's constant times the frequency, and the frequency from the speed of light over the wavelength. So the two constants arrive together as hc . That is 1.99×10^{-25} joule metres, and worth remembering. Dividing a power by the energy of one photon is what turns watts into a count. And the count is what the receiver's limit is really about.

TAKEAWAY 23W

A photon's energy is fixed by its wavelength alone, so a power in watts is a rate of arrivals in photons a second.

5.2 Six thousand a bit

RECEIVER LAB • A PERSON • BALLPARK

SCENE 31W

Duarte has the receiver's sensitivity on the bench in watts, the line rate beside it, and this morning's photon energy on a slip of paper. She wants the arrivals per bit.

GUIDE 65W

Four quantities are offered and three of them make the answer. Divide a power by the energy of one arrival and you have arrivals a second. Divide again by the bits a second and you have arrivals a bit. The launch power is on the list because it belongs at the other end of the span. It is 151 times larger than what arrives here.

ASK 11W

How many photons arrive in one bit at the receiver's sensitivity?

TAKES AS READ

a stream of arrivals has a spread of about the square root of its count • photons: energy, momentum and the quantum of light — taken as read

VERDICT 85W

About six thousand photons arrive in each bit. That is a large number of anything except a count. The spread of a count is about its square root. So six thousand arrivals carry a natural unevenness of about eighty, which is over one per cent. No improvement to the electronics touches it, because it is a property of light arriving in pieces. The two ways to get further are more photons — a better amplifier or a shorter span — or fewer bits a second.

TAKEAWAY 26W

A receiver at the end of a long span is counting a few thousand arrivals a bit, and the root of that count is its floor.

5.3 Wider than it left

CABLE TERMINATION ROOM · A ROOM · CHOICE

SCENE 34W

Nilsen has the launched pulse and the received one on the same screen, the second visibly broader than the first. Nothing about the span has changed except that the light has been along it.

GUIDE 57W

Nilsen has two traces of one pulse, with eighty-four kilometres of glass between them. Broadening and weakening are different faults with different cures. Read each option and ask which of the two it would produce. Then ask where in the span it would happen: in the glass, at the joints, or in the equipment at either end.

ASK 9W

Why does the pulse arrive wider than it left?

TAKES AS READ

a pulse of light contains a small range of wavelengths rather than one exactly

VERDICT 89W

Because the pulse is not one wavelength but a narrow band of them. And the index of the glass is slightly different for each. So the parts of the pulse arrive at slightly different times, and it spreads. That is dispersion, a property of the material. It is why a long span has a bit rate limit as well as a power limit. It is also why the fibre's own index is not quite the number a pulse travels at. That is the correction the ship's chart is missing.

TAKEAWAY 17W

A pulse is not one wavelength, and anything that depends on wavelength therefore acts on its shape.

5.4 What came back, and when — Review

TEST ROOM • A ROOM • CALLBACK • PROTOCOL

SCENE 40W

Tobias Meier, the test room technician, has four readings from the diverse route pinned on the board. None of them came off the same instrument, and none of them has been carried any further than the box that produced it.

GUIDE 51W

Four readings and four descriptions. Sort them by what the instrument itself could know. A quantity a box measured, a quantity software has already converted and a note of the conditions are three different kinds of thing. Only the first of the three is evidence before anybody has argued about it.

ASK 11W

Match each reading from the diverse route to what it is.

BEHIND THE BUTTON 93W

Why this is a matching board rather than four separate questions. Each description is written to be defensible against more than one of the readings, so choosing them one at a time lets a nearly-right answer through. Placing all four forces the comparison the stop is about.

What a condition is doing on a list of measurements. It is not evidence about the cable at all. It is the thing that decides whether two readings taken on different nights are comparable, and a reading whose conditions are unrecorded cannot be compared with anything.

TAKES AS READ

a reflectometer reports the delay of a pulse that has been out and back • the position printed beside a trace is worked out from the delay by software • time of flight, and turning a delay into a distance — taken as read • power in decibels, and why a loss budget adds — taken as read

VERDICT 90W

Only one of the four is a number nobody has interpreted. The delay is what the instrument recorded, so it carries no assumption at all. The power is a measurement too, and it becomes a margin only once an allowance is subtracted from it. The printed position is a conversion already performed, resting on a speed in the glass and a slack figure from a lay chart eleven years old. The pulse width is none of those. It is a condition, and conditions decide whether two readings may be compared.

TAKEAWAY 24W

A set of readings is usually three different kinds of thing, and saying which kind each one is comes before comparing any of them.

END OF THE DAY 16W

Light arrives in pieces of a fixed size, so a power is a rate of arrivals.

One electron for one photon

PLAN CARD 138W

Monday, and seven working days to the ship. Karl Osgood, the photodiode technician, has measured the receiver's diode himself. He gets 82 arrivals converted in every hundred. The data sheet claims ninety. Duarte needs the current per watt either way. The whole sensitivity figure rests on it, and the loss budget was closed against that figure. Ramnarine has four defective joints from the shore end on her bench. She wants to know which of them a trace could have seen. Today you derive the current that one watt of light produces. You also sort the four faults by the mark each one leaves. Then you settle with Sam Achebe, the fibre preparation technician, how two reflections can cancel each other out. A sensitivity figure ten per cent too hopeful is a decibel of margin that is not there.

BRIEFING 13W

The photodiode's data sheet gives a number and Osgood measures a different one.

WHAT YOU DECIDE 15W

Get the receiver's current per watt out of the physics rather than off a label.

6.1 One amp a watt, near enough

RECEIVER LAB · A ROOM · DERIVE

SCENE 30W

Osgood has the diode's measured efficiency on the bench — 82 charges for every hundred arrivals — and this morning's photon energy beside it. Duarte wants the current per watt.

GUIDE 60W

You build the working a line at a time. A watt is a number of arrivals a second. Each arrival that is converted releases one charge. So the current is a count times the charge on one of them. Take the arrivals first, apply the efficiency, then attach the charge. The wavelength enters only through the energy of one arrival.

ASK 12W

What current does one watt at this wavelength produce in this diode?

BEHIND THE BUTTON 173W

Why the answer is close to one. At this wavelength a photon carries 0.80 electron volts. An electron volt is exactly the energy an electron gains through a volt. So the current per watt comes out near the reciprocal of that number of volts. An efficiency of 0.8 and a photon energy of 0.8 electron volts happen to cancel. That is a coincidence, and a useful one to notice.

Why the responsivity rises with wavelength. Longer wavelengths mean less energy per arrival, so a watt is more arrivals, so more charges: the responsivity of a perfect detector is proportional to the wavelength. That continues only until the arrivals are too weak to free a charge at all, which is the cut-off.

What the ten per cent is worth. The difference between the sheet's 0.90 and the measured 0.82 is 0.4 decibels of sensitivity. Against 2.2 decibels of margin, that is a fifth of everything the span has spare. It is also the difference between a figure somebody measured and a figure somebody printed.

TAKES AS READ

a converted arrival releases exactly one unit of charge · the photoelectric effect, and a threshold that is about energy per photon — taken as read

VERDICT 87W

One watt produces about 1.03 amps. A watt at 1550 nanometres is 7.8×10^{18} arrivals a second. Of those, 82 per cent free a charge. Each charge is 1.602×10^{-19} coulombs. The product is a current, and dividing by the watt makes it a responsivity. Written as one expression it is the efficiency times the charge over the photon energy. That shows the wavelength dependence. At 1300 nanometres the same diode would give 0.86 amps a watt, with no change to the diode at all.

TAKEAWAY 20W

Responsivity is the efficiency times the charge divided by the energy of one arrival, and it grows with the wavelength.

6.2 Four faults, four traces

SPLICE TRAILER · A PERSON · PROTOCOL

SCENE 33W

Ramnarine has four faults on the bench, all recovered from shore ends over the years, and a trace beside each. Every one of them lost light and no two of them look alike.

GUIDE 63W

Each fault on the left has one signature on the right. What separates them is whether the light that is lost comes back towards the instrument or leaves the fibre altogether. It also matters whether the loss is at one point or spread along a length. Two of the four look identical in how much they lose and nothing alike in what returns.

ASK 11W

Match each fault to the signature it leaves on a trace.

TAKES AS READ

an instrument sees only the light that returns to it

VERDICT 96W

A break reflects about 3.6 per cent at the glass-air step, which is a tall spike, and nothing continues past it. A bend has no index step at all, so light leaves the core and none comes back: a step with nothing on it. A dried joint has a step in index but a small one, so a small spike and the trace goes on. And a crushed length loses gradually rather than at a point, which shows as the slope getting steeper for twenty metres. The step heights alone would not separate any of these.

TAKEAWAY 28W

Two faults can lose the same light and be told apart by what comes back, which is why a trace has a shape as well as a step.

6.3 Sixty-five thousand traces

TEST ROOM · A ROOM · CHOICE

SCENE 23W

Meier has the instrument set to average sixty-five thousand traces, which takes eleven minutes. The same measurement with a thousand takes ten seconds.

GUIDE 54W

Meier is spending eleven minutes on one measurement instead of ten seconds. He wants to know what he is buying. Some of what a trace shows changes from sweep to sweep. Some of it is identical every time. Ask of each option which kind it names, and whether repeating the measurement could touch it.

ASK 7W

What does averaging sixty-five thousand traces buy?

TAKES AS READ

independent measurements of the same thing average towards it

VERDICT 82W

It reduces the part of the trace that varies from one pulse to the next. It does so as the square root of the number averaged. Sixty-five times more traces than a thousand buys a factor of eight, which is nine decibels. Sixty-four times the time for nine decibels is the bargain on offer. What it cannot touch is anything the same in every trace. A reflection from a connector, or the instrument's own dead zone. Averaging identical things returns the thing.

TAKEAWAY 21W

Averaging reduces what varies between measurements as the root of their number, and does nothing at all to what does not.

6.4 Current, and not light — Review

AMPLIFIER AND POWER HALL · A ROOM · CALLBACK · SEQUENCE

SCENE 38W

Bruno Salas, the power feed engineer, has a wall chart of the erbium levels taped up beside repeater 6's telemetry. Four things happen to one atom in that coil, and the chart has them in no particular order.

GUIDE 44W

Four cards, and the rail here is a clock. Start from the fact that nothing can be amplified until an atom is already waiting in the long-lived level. Ask of each card whether it puts an atom there, keeps it there, or spends it.

ASK 11W

Order the four events in the life of one erbium atom.

BEHIND THE BUTTON 108W

Why one of the four levels is short-lived and one is not. The pump lands atoms on a level they leave almost at once, and they leave it downwards, to a level they sit on for milliseconds. Only a level atoms sit on can hold a population, which is why the pump wavelength and the signal wavelength are different.

What this chain has to do with a drive current. Nothing on the telemetry screen touches any of the four steps. A current says power is going into the pump diode. How many atoms that power lands on the upper level is a question about the diode's optical output.

TAKES AS READ

an amplifier works by holding more atoms excited than not · atomic energy levels, and the wavelengths they allow — taken as read

VERDICT 88W

Amplification is a queue. The pump lifts atoms to a level they do not stay in. They settle almost at once to a longer-lived level below it, and a population can be held there. An arriving signal photon then finds an atom already excited and triggers a second photon exactly like itself. Nothing in that chain is reached by a drive current: the pump's optical output sets how many atoms are waiting. Everything the amplifier delivers is a count of atoms rather than a reading on a supply.

TAKEAWAY 24W

Gain is a count of atoms already waiting in the upper level, so every step that fills that level is part of the answer.

END OF THE DAY 20W

A detector converts arrivals to charges, so its current per watt is fixed by the wavelength and its own efficiency.

Through forty millimetres of steel

PLAN CARD 128W

Tuesday. Wherever the ship is sent, it will be putting a new repeater housing into the water. Farouk cannot pass a weld she cannot see. What her film sees is whatever gets through forty millimetres of steel. Joe Whitlock, the radiation safety officer, sets the barrier before anyone goes near the bay. Sarraf meanwhile has four possible causes and a week to choose between them. Today you work out how much of a gamma beam a housing lets through. You also put the four causes in the order they should be tested in. Then you settle with Sam Achebe how two reflections can cancel. The housing has to be signed off before the ship sails. That is true whether or not anybody has agreed where it is going.

BRIEFING 16W

The ship needs a housing radiographed before it sails, and the source is eight months old.

WHAT YOU DECIDE 16W

Work out what gets through a housing, and settle what a trace can and cannot separate.

7.1 Seven per cent, and no more

RADIOGRAPHY BAY · A ROOM · DERIVE

SCENE 35W

The housing wall is forty millimetres of steel and the source's beam is attenuated at 0.666 per centimetre in it. Farouk wants the fraction that reaches the film, and the thickness that would halve it.

GUIDE 54W

You build the working a line at a time. Attenuation removes a fixed fraction per unit thickness, so the fraction remaining is an exponential in the thickness. Get that fraction first, then invert the same relation to find the thickness that halves the beam, then count how many of those the wall is worth.

ASK 12W

What fraction of the beam reaches the film through the housing wall?

BEHIND THE BUTTON 153W

Why a half-value thickness is the useful form. A material has no thickness that stops a gamma beam; it has a thickness that halves it, and another one that halves it again. So shielding is quoted as that thickness, and a barrier is designed by counting how many of them are in the way.

Why the exponent and the count agree. Forty millimetres at 0.666 per centimetre is an exponent of 2.66. Divide 2.66 by the natural logarithm of two and you get 3.85. So the wall is 3.85 half-value thicknesses, and two to the power of minus 3.85 is 0.069. The two routes are the same statement written in different bases.

What it costs Farouk. Seven per cent through means an exposure fourteen times longer than a film with nothing in the way, before any allowance for the source being eight months old. That second factor arrives tomorrow, and the two multiply.

TAKES AS READ

equal thicknesses of a material remove equal fractions of a beam · radioactive decay as a random process with a fixed chance — taken as read

VERDICT 86W

About seven per cent gets through. The fraction remaining is e to the minus the coefficient times the thickness, which is e to the minus 2.66. The same coefficient gives a half-value thickness of 1.04 centimetres, and the wall is 3.85 of those. And two to the minus 3.85 is the same 0.069. The two forms are one relation in different bases. Neither form contains a thickness that stops the beam. That is why a barrier is a distance as often as it is a slab.

TAKEAWAY 17W

A half-value thickness is the natural way to state shielding, because attenuation is exponential and never complete.

7.2 Four causes, one week

AMPLIFIER AND POWER HALL · A PERSON · SEQUENCE

SCENE 30W

Four things could have taken four decibels out of this span overnight, and there are five working days left. Tanaka and Sarraf have agreed the list and not the order.

GUIDE 61W

Four tests, and the rail is not a clock. All four could be started this morning. Put them in order of what each one costs before it answers, cheapest first. Ask of each whether it needs the ship. Ask whether it needs the traffic moved off. Ask whether it can be done from this building with the span as it stands.

ASK 11W

Order the four tests by what each costs before it answers.

TAKES AS READ

a test that costs nothing to try should be tried before one that costs a week

VERDICT 94W

Re-reading traces already in the drawer costs an afternoon and eleven years of them exist. A fresh trace with a shorter pulse costs an hour of the instrument and a technician. Switching to the spare pump is a real intervention on a live repeater and cannot be undone from shore, so it costs a decision. Sending the ship costs the charter, at forty thousand a day, and commits everybody to a position. Two of those four would have answered the question, and the order matters because the cheap ones make the expensive one unnecessary.

TAKEAWAY 19W

Order tests by what they cost before they answer, because the cheap ones often make the expensive ones unnecessary.

7.3 Two reflections, and a gap between them

SPLICE TRAILER · A ROOM · CHOICE

SCENE 34W

Achebe has a connector face under the microscope with a coating on it, and the same face uncoated beside it. The coated one reflects visibly less and neither is polished better than the other.

GUIDE 52W

Achebe has two faces under the microscope, and neither is polished better than the other. Light can be absorbed, redirected, or made to cancel itself out. Those are three different things, and only one of them leaves the light in the fibre. Ask of each option which of the three it describes.

ASK 11W

How does a coating on a connector face reduce its reflection?

TAKES AS READ

two waves arriving out of step by half a wavelength cancel

VERDICT 84W

The coating makes two reflections instead of one, from its front face and its back face. If the extra path between them is half a wavelength, the two arrive out of step and cancel. Nothing is absorbed: the light that would have come back goes forward instead. That is why the thickness matters, and why it is a quarter of a wavelength rather than any old thin layer. It is the same arithmetic as the gel in a mechanical joint, reached a different way.

TAKEAWAY 21W

Two reflections separated by the right path difference arrive out of step and cancel, which removes a reflection without absorbing anything.

END OF THE DAY 17W

Equal thicknesses take equal fractions, so shielding is a thickness that halves rather than one that stops.

A coating that cancels

BEFORE THE DAY — FOLLOW

The duct route, walked by the man who terminated it

Halvard Nilsen, the cable termination technician, walks the duct from the beach manhole to the termination room the same way every week and the route is not on the drawing at the accuracy that matters. Stay with him along it. Where he stops is where the duct is shallow.

PLAN CARD 127W

Wednesday. Meier has found something in the drawer. The reflection off the instrument's own front connector is bigger than the spike in the trace. It has been there for eleven years. Ramnarine can coat a face to get rid of it. She wants the thickness derived here, not ordered out of a catalogue. Osgood has a diode on the bench that works well at this wavelength. At two thousand nanometres it does nothing at all. Nobody in the building can say why. Today you design the coating. You also settle where a detector's cut off comes from. Then you look with Okonkwo at what a bend does that a break does not. Every trace on this station has the instrument's own reflection taken off it by hand.

BRIEFING 14W

Every connector in the building reflects, and one of them is inside the instrument.

WHAT YOU DECIDE 14W

Design the film that removes a reflection, and find where a detector stops working.

8.1 Three hundred and twenty nanometres

SPLICE TRAILER · A ROOM · DERIVE

SCENE 37W

Ramnarine can put a coating on a connector face to any thickness and to a choice of indices. The face is glass at 1.4682, the light is at 1550 nanometres, and air is on the other side.

GUIDE 63W

You build the working a line at a time. A coating makes two reflections, one at each of its faces, and the question is what thickness puts them exactly out of step. Count the path difference in the film. Remember that the wave travels more slowly in it. Then ask what index makes the two reflections the same size, because cancelling needs both.

ASK 12W

How thick should the coating be, and what index should it have?

BEHIND THE BUTTON 152W

Why the film is a quarter of a wavelength and not a half. The extra path is twice the thickness, because the second reflection goes down and back, so a quarter-wavelength film gives a half-wavelength path difference. That is the difference between a coating that cancels and one that does nothing at all.

Why the wavelength inside the film is the one that counts. The wave is slower in the film, so its wavelength there is the vacuum wavelength divided by the film's index. The thickness that matters is therefore λ over four n , not λ over four.

Why the index is a geometric mean. Cancelling needs the two reflections equal as well as opposite. The two step sizes are equal when the film's index is the root of the product of the two either side. That is 1.212 here, which is close to magnesium fluoride, which is what is actually used.

TAKES AS READ

a wave reflected at each face of a film travels an extra distance of twice the thickness
• snell's law at a boundary — taken as read

VERDICT 95W

The film should be about 320 nanometres thick with an index of 1.212. Two reflections leave it, and the second travels twice the thickness further. So a thickness of a quarter of the wavelength inside the film puts them half a wavelength apart, and they cancel. The wavelength inside the film is the vacuum wavelength divided by the film's index, which is why the thickness depends on that index as well. And they only cancel completely if they are the same size, which happens when the film's index is the geometric mean of its neighbours'.

TAKEAWAY 20W

A coating cancels a reflection when its thickness gives a half-wavelength path difference and its index equalises the two reflections.

8.2 Bright and useless

RECEIVER LAB · A PERSON · CHOICE

SCENE 32W

Osgood has two sources on the bench and one diode. A dim source at 1550 nanometres gives a clean current. A source twenty times brighter at 2000 nanometres gives nothing at all.

GUIDE 52W

Osgood has one diode and two sources, and the brighter of the two does nothing at all. How much light arrives and what each arrival carries are separate quantities. Turning the brightness up changes the first and leaves the second alone. Ask of each option which of the two it depends on.

ASK 12W

Why does the brighter source at the longer wavelength give no current?

TAKES AS READ

a charge is freed by one arrival rather than by several together

VERDICT 88W

Each arrival at 2000 nanometres carries 0.62 electron volts, and the diode's material needs 0.75 to free a charge. Brightness is the number of arrivals. Arrivals that individually cannot free a charge do not free one between them. So twenty times as many of them still gives nothing. That is the whole content of a threshold. It is about the energy of one piece of light, which depends only on the wavelength. That is the reason a detector is chosen for a band rather than for a power.

TAKEAWAY 17W

A threshold in wavelength is a statement about energy per arrival, so brightness cannot get past it.

8.3 Four bends, and what each one costs

CABLE TERMINATION ROOM · A ROOM · CASEBOOK

SCENE 36W

Okonkwo has four offcuts down off the rack in the ship's store, each bent round a different radius, with the loss of each written beside it. None of them is damaged and all four lose light.

GUIDE 54W

Each observation on the left has one explanation on the right. What separates them is what happens to the angle a ray makes with the core boundary when the fibre is bent. The condition for staying in the core is an angle rather than a strength. A bend changes angles without changing anything else.

ASK 11W

Match each observation about a bent fibre to what explains it.

TAKES AS READ

light stays in a core only while it meets the boundary beyond the critical angle

VERDICT 96W

A bend does not damage anything; it changes the angles. On the outside of a tight bend the boundary is tilted relative to the rays. Some of them then arrive inside the critical angle and refract out instead of reflecting. A gentle bend tilts it too little to matter. Longer wavelengths spread further into the cladding and so leave first, which is exactly the test that separates a bend from a break. And a bend has no index step anywhere in it, so nothing comes back — which is the other half of the same test.

TAKEAWAY 19W

A bend changes the angle a ray meets the boundary at, so light leaves without anything having been damaged.

END OF THE DAY 11W

A quarter-wavelength film cancels a reflection by interference, without absorbing anything.

A tenth of what it was

PLAN CARD 113W

Thursday. The housing radiograph is now on the critical path. Farouk's source was renewed eight months ago. Her exposures have grown longer every month since. Whitlock will not open the bay until the barrier distance is worked out for the exposure actually being made. Tanaka has spent the morning on the pump's two wavelengths. She has a number she wants checked. Today you work out what the source is worth now. You also set the barrier for it. Then you settle where the gap between the pump's energy and the signal's energy goes. If the exposure runs past six in the evening the bay cannot be used. The housing then waits another day.

BRIEFING 16W

The housing has to be signed off before the ship sails, and the exposures keep growing.

WHAT YOU DECIDE 17W

Work out what eight months has done to the source, and how far back the barrier goes.

9.1 Two hundred and forty-three days

RADIOGRAPHY BAY · A ROOM · DERIVE

SCENE 31W

The source has a half-life of 73.8 days and was renewed 243 days ago. Farouk wants what fraction of its original strength is left and what that does to her exposures.

GUIDE 58W

You build the working a line at a time. A half-life and a decay constant are two ways of writing the same fact. The exponential is the same shape as yesterday's attenuation, with time in place of thickness. Get the constant, then the fraction, then check it by counting half-lives — because the two routes have to agree.

ASK 12W

What fraction of the source's original activity is left after 243 days?

BEHIND THE BUTTON 170W

Why the two forms agree. e to the minus the constant times the time, and two to the minus the number of half-lives, are the same function written in different bases. The logarithm of two is the conversion between them. Being able to move between them is what makes a decay estimable in the head.

Why the exposure factor is the reciprocal. A film needs a certain number of arrivals whatever the rate they come at, so a source a tenth as strong needs ten times as long. Both factors multiply: a tenth of the source and seven per cent through the steel is a hundred and forty times a bare film's exposure.

What the date beyond which is. At 73.8 days a half-life, this source loses another half every two and a half months, and the bay's working day sets a ceiling on the exposure. That ceiling arrives about eleven months in, which is why the renewal interval is what it is rather than being a matter of policy.

TAKES AS READ

each nucleus has a fixed chance of decaying in the next second, whatever its history · radioactive decay as a random process with a fixed chance — taken as read

VERDICT 83W

About a tenth is left. The decay constant is the logarithm of two divided by the half-life, which is 0.00939 per day. The fraction remaining is e to the minus that times 243 days. The check is the other route: 243 days is 3.29 half-lives, and two to the minus 3.29 is the same 0.102. Both are the same statement about a nucleus having a fixed chance of decaying per second, and neither depends on how much source there was to start with.

TAKEAWAY 17W

A decay constant and a half-life are the same fact, and a source's schedule follows from either.

9.2 Eight metres, or thirty-two

RADIOGRAPHY BAY • A PERSON • BALLPARK

SCENE 34W

Whitlock has the dose rate measured at two metres from the pot and a limit for the boundary of the controlled area. He wants the distance at which the rate is inside the limit.

GUIDE 63W

Four quantities are offered and three of them make the answer. A small source spreads its output over a sphere, so the rate falls as the square of the distance. The distance needed is the reference distance times the root of the ratio of the rates. The exposure time is on the list because it is not part of a rate at all.

ASK 10W

How far back does the barrier go for today's exposure?

TAKES AS READ

a small source spreads its output evenly over a sphere • radioactive decay as a random process with a fixed chance — taken as read

VERDICT 83W

The barrier goes to about 18.5 metres. Intensity from a small source falls as the inverse square of the distance. So reducing a rate by a factor of 85 needs a distance greater by the square root of 85, which is 9.2. That is why distance is the cheap answer to a strong source and lead is the expensive one. It is also why a collimated beam breaks the arithmetic entirely. A beam that has been aimed is not spreading over a sphere.

TAKEAWAY 25W

Intensity from a small source falls as the square of distance, so the barrier is set by a square root rather than by a ratio.

9.3 Two wavelengths, one atom

AMPLIFIER AND POWER HALL · A ROOM · CHOICE

SCENE 40W

Tanaka has the two wavelengths on the board: the pump at 980 nanometres and the signal at 1550. Every pump arrival that does its job ends up as a signal arrival, and the two do not carry the same energy.

GUIDE 51W

Tanaka has the two wavelengths on the board. A pump arrival at 980 nanometres carries more energy than a signal arrival at 1550. Energy is one book and the count of arrivals is another. Ask of each option whether it keeps both books straight. Then ask where it puts the difference.

ASK 13W

What becomes of the difference between a pump arrival and a signal arrival?

TAKES AS READ

an excited atom can be made to release its energy into a passing wave

VERDICT 91W

It ends up as heat in the glass. A pump arrival at 980 nanometres carries 1.27 electron volts. It lifts an erbium atom to a state it drops from, quickly and silently, to the level the signal works from. That drop is 0.47 electron volts that never becomes light. What is conserved is the count: one pump arrival can produce at most one signal arrival. That ceiling is what sets an amplifier's maximum output at the pump power times the ratio of the two wavelengths, whatever else is done to it.

TAKEAWAY 20W

A pump arrival is more energetic than the signal arrival it produces, and the difference has to leave as heat.

END OF THE DAY 22W

A source falls by a fixed fraction in a fixed time, so every exposure it is used for has a schedule attached.

The kilometre the trace is out by

PLAN CARD 139W

Friday, and the chart goes to the ship on Monday. Okonkwo has the lay record open at the page he has been pointing at for a fortnight. It says 1.5 per cent of slack was paid out over the route. The fibres are wound in a spiral inside the cable. So there is more fibre than there is cable. Meier has the instrument's index setting. It is the core's index, not the one a pulse travels at. Tanaka has permission to switch repeater 6 to its spare pump this afternoon. Today you correct the distance for all of it. You also predict what the spare pump will do, and then measure it. Then you settle with Duarte which description of light a receiver needs. The grapnel window is half a kilometre wide, and the correction is larger than that.

BRIEFING 17W

Okonkwo has been saying an optical distance is not a position, and today it is a number.

WHAT YOU DECIDE 15W

Correct the distance for what a pulse actually travels at, and for the cable's slack.

10.1 Eighty-two point nine

TEST ROOM · A ROOM · DERIVE

SCENE 36W

The instrument reported 84.6 kilometres using the core's index of 1.4682. A pulse travels at the group index, 1.4718. There is 0.3 per cent more fibre than cable, and 1.5 per cent more cable than route.

GUIDE 72W

You build the working a line at a time. Three corrections, each a small ratio, and each one in a definite direction. The instrument used too small an index. The fibre is longer than the cable it is in. The cable is longer than the route it lies on. Get the sign of each right before touching the arithmetic, because two of the three go the same way and one does not.

ASK 13W

How far along the route is the event, once every correction is applied?

BEHIND THE BUTTON 186W

Why the group index is the right one. A pulse is a band of wavelengths, and what travels at c over the core's index is a single wavelength. The pulse — the thing whose arrival is timed — travels a little more slowly, at c over the group index. So the same delay corresponds to a slightly shorter distance.

Why the fibre is longer than the cable. The fibres are laid up helically round the cable's core so that bending the cable does not stretch them. So a metre of cable contains slightly more than a metre of fibre. It is a fixed property of the cable's construction and it is on the maker's data sheet.

Why the slack is the largest term and the least certain. A cable is paid out faster than the ship advances so that it lies on the bottom rather than bridging the high spots. 1.5 per cent over 84 kilometres is 1.25 kilometres. It is an average over a route whose slack was not uniform. That is exactly why Okonkwo will not offer a position without a tolerance on it.

TAKES AS READ

a cable laid with slack is longer than the route it lies along

VERDICT 87W

The event is about 82.9 kilometres along the route, 1.7 kilometres closer than the trace reads. The instrument's index was too small, so its distance was too large: correcting it takes off 207 metres. The fibre is 0.3 per cent longer than the cable, and the cable 1.5 per cent longer than the route. Both of those are divisions rather than multiplications. The total is more than three times the grapnel window, which is what makes it the difference between finding the fault and dragging past it.

TAKEAWAY 20W

Three ratios separate an optical distance from a position, and all three have to be applied with their own signs.

10.2 Three and a half decibels, or nothing

AMPLIFIER AND POWER HALL · A ROOM · VERIFY

SCENE 34W

Repeater 6 has a spare pump laser that has never been used. Tanaka can switch to it from shore, once, and the receiver's power meter is the only instrument that will see the result.

GUIDE 61W

Lock a prediction for how much the received power will rise before the switch is made. Nothing about it can be changed afterwards, which is the point of making it first. Then switch, and then spend the measurement. A switch that is confirmed to have happened and never measured establishes that the relay works. It establishes nothing whatever about the pump.

ASK 15W

Predict the rise in received power, switch to the spare pump, and report what happened.

TAKES AS READ

an amplifier's gain follows the optical power of its pump

VERDICT 98W

The received power rises about 3.6 decibels. An amplifier's gain follows its pump's optical power. A pump that has aged puts out less at the same current. If the original is down to 40 per cent of the 118 milliwatts it was logged at, a fresh one restores a factor of two and a half. That is 4 decibels of gain, less what the amplifier itself gives back. That is most of the 4.1 decibels the span lost. The failure specific to this decision is finishing without measuring: the telemetry will confirm the switch whatever the light does.

TAKEAWAY 23W

A prediction locked before an intervention is a test of the model; the same number chosen afterwards is a description of the outcome.

10.3 Both, on the same bench

RECEIVER LAB · A PERSON · CHOICE

SCENE 29W

Duarte has two instruments running on the same fibre. One shows an interference pattern from two paths recombined. The other counts arrivals, one at a time, on a diode.

GUIDE 50W

Duarte has two instruments running on one fibre, and two results that look incompatible. One shows a pattern that only a wave makes. The other counts arrivals one at a time. Ask of each option whether it decides what light is, or whether it says which description a measurement needs.

ASK 8W

How should the two results be held together?

TAKES AS READ

a description of light is chosen for a measurement rather than for all time

VERDICT 92W

Each measurement needs the description that fits what it does. The interference pattern exists because two paths recombine with a path difference, which is a statement about a wave. The arrivals are counted one at a time on a diode, which is a statement about quanta. Both are true of the same light on the same fibre. What is not available is a picture in which light is only one of them. A receiver's design needs both at once: a wavelength to be filtered, and a count to set the error floor.

TAKEAWAY 18W

A model is chosen for a purpose, and two incompatible-looking pictures can both be needed on one bench.

END OF THE DAY 21W

An optical distance becomes a position only after the pulse's own speed and the cable's slack are taken out of it.

What the pump could ever give

PLAN CARD 136W

Monday, and two working days to the ship. Friday's switch gave back 3.6 of the 4.1 decibels the span had lost. That is a fact about repeater 6, not about the seabed. Tanaka wants the ceiling derived. The ceiling is the most signal power a pump of a given size could ever produce. With it, the old pump's decline can be stated as a number instead of guessed at. Sarraf has every record of the last three weeks laid out on one table. Today you take the ceiling on the amplifier's output. You also decide which of the four causes accounts for all of it. Then you settle with Ramnarine how the two wavelength comparison is made at all. Whatever goes on the chart on Wednesday is what forty thousand a day is spent looking for.

BRIEFING 17W

The spare pump gave back most of what the span lost, and the fibre was never damaged.

WHAT YOU DECIDE 14W

Put a ceiling on the amplifier's output, and decide which cause fits every record.

11.1 Seventy-four milliwatts, at most

AMPLIFIER AND POWER HALL · A ROOM · DERIVE

SCENE 39W

The pump is at 980 nanometres and the signal at 1550. Repeater 6's pump was logged at 118 milliwatts when it was fitted four years ago. Tanaka wants the most signal power that could ever come out of it.

GUIDE 71W

You build the working a line at a time. The ceiling is a count: each pump arrival can end up as at most one signal arrival, and the signal arrival carries less energy. So the largest possible added signal power is the pump power times the ratio of the two energies. That is the ratio of the two wavelengths the other way up. Then compare it with what the span lost.

ASK 12W

What is the most signal power 118 mW of pump could add?

BEHIND THE BUTTON 172W

Why the ceiling is a ratio of wavelengths. A pump arrival at 980 nanometres carries hc over 980; a signal arrival carries hc over 1550. So the same number of arrivals a second carries 980 over 1550 as much power, which is 0.632. 118 milliwatts of pump can therefore add at most 74.6 milliwatts of signal.

Why nothing gets past it. Every route from pump to signal goes through one excited atom releasing one arrival. There is no mechanism in this glass by which one pump arrival produces two signal arrivals. So the ceiling is a statement about counting rather than about engineering. A better amplifier design approaches it rather than exceeding it.

What it says about repeater 6. A pump down to 46 milliwatts has a ceiling of 29 milliwatts against the original's 74.6. That is 4.1 decibels of gain lost. It is the number the span lost in one night, arrived at from the pump's own physics rather than from the trace. That is the end of the fortnight's argument.

TAKES AS READ

each pump arrival can produce at most one signal arrival · atomic energy levels, and the wavelengths they allow — taken as read

VERDICT 90W

At most 74.6 milliwatts. The pump's power is a number of arrivals a second, each carrying 1.27 electron volts. Every one of them can produce at most one signal arrival carrying 0.80. So the power that can be added is the pump power times the ratio of those energies, which is the ratio of the wavelengths inverted. Applying the same ceiling to a pump that has fallen to 46 milliwatts gives 29. The difference between the two ceilings is 4.1 decibels, which is what the span lost in a night.

TAKEAWAY 26W

The most an amplifier can add is its pump power scaled by the ratio of the two wavelengths, because arrivals are conserved and energies are not.

11.2 Four records, one cause

TEST ROOM · A PERSON · DIAGNOSIS

SCENE 27W

Everything from three weeks is on the table: the trace with its step, the two-wavelength comparison, the power feed's log, and Friday's switch to the spare pump.

GUIDE 59W

Look at every record before choosing, including the two that are not alarming. The power feed is in order. The two-wavelength comparison shows the same loss at both, and that is the record that does most of the work here. A bend loses more at the longer wavelength. A repeater's gain does not care which wavelength it is amplifying.

ASK 7W

Which cause accounts for all four records?

TAKES AS READ

a bend loses more at longer wavelengths and an amplifier's gain does not

VERDICT 113W

Three of the four records point the same way once the quiet ones are read. A bend and a crush both lose more at 1625 nanometres than at 1550. The comparison shows the same loss at both. That rules out anything that works by light leaving the core. A dried joint would have put a reflection on the step, and would not have been mended by switching a pump three kilometres further out. The pump's decline accounts for a loss equal at both wavelengths, at the position of the repeater, with the feed in order. It also accounts for 3.6 of the 4.1 decibels coming back when a fresh pump was switched in.

TAKEAWAY 26W

A cause has to account for the quiet records as well as the loud one, and the quiet ones are usually what rule the others out.

11.3 One fibre, two colours

SPLICE TRAILER · A ROOM · CHOICE

SCENE 34W

The two-wavelength comparison is the record that settled today. It is made inside the instrument's own head, on a glass plate ruled with hundreds of fine lines. Ramnarine has never had that head open.

GUIDE 71W

Ramnarine has never had that head open, and she wants to know what is doing the work in it. A diffraction grating — the ruled plate — is not a filter, and the two do different jobs. Separating two wavelengths means both come out, in different places. Selecting one means the other is gone. Ask of each option which of those it describes, and what physical thing does the telling apart.

ASK 9W

How does a grating separate two wavelengths arriving together?

TAKES AS READ

waves from many evenly spaced sources reinforce only in particular directions

VERDICT 82W

Light from each of the grating's many lines travels a slightly different distance to any given direction. The directions where all of them reinforce depend on the wavelength. So each wavelength leaves at its own angle, and the two arrive at different places on a detector: separation rather than selection. A filter would pass one and throw the other away. That is a different instrument and a different measurement. You would have to make it twice and hope nothing changed in between.

TAKEAWAY 15W

Separating a mixture and selecting one part of it are different operations with different costs.

END OF THE DAY 21W

One pump arrival can make at most one signal arrival, which puts a ceiling on gain that no design gets past.

What the ship is told

PLAN CARD 122W

Tuesday, and the ship sails in the morning. The fortnight has settled the old argument. Sarraf was right to act on the trace. Okonkwo was right about what the trace was saying. The loss sits at repeater 6, in the pump rather than in the glass. The position on the chart was out by 1.7 kilometres, for three reasons that all pointed the same way. Farouk has a housing radiographed and signed. Whatever is on the chart at four o'clock is what forty thousand a day goes looking for. Today you close the loss budget against the repeater. You also spend what is left of the spares money. Then you settle with Tanaka what the ship is actually being sent to do.

BRIEFING 13W

The chart goes out this afternoon and the ship sails in the morning.

WHAT YOU DECIDE 15W

Close the loss budget against the repeater, and decide what the charter is spent on.

12.1 Four decibels, accounted for

TEST ROOM · A ROOM · DERIVE

SCENE 36W

The budget from day four is on the board: 21.8 decibels total, of which 4.1 arrived in one night. Friday's switch returned 3.6. Sarraf wants the whole thing closed, term by term, with nothing left over.

GUIDE 67W

You build the working a line at a time. Take the budget as it stood before the event, add what the pump's decline accounts for, and see what is left unexplained. A budget that closes to within a tenth of a decibel is a different kind of statement from one that closes to within a decibel. The difference decides what the ship is told to look for.

ASK 13W

How much of the 4.1 dB is accounted for, and what is left?

BEHIND THE BUTTON 163W

Why the residue matters more than the total. Half a decibel unaccounted for could be a second cause. It could be the amplifier giving slightly less gain at the higher input. It could be the measurement. Saying which is what the last two days have been for, and saying it is not the same as having no residue at all.

Why the ship still sails. A pump on the seabed cannot be replaced from shore. The spare is now the working pump and there is no second spare. So repeater 6 has one failure left in it before the span goes down, with the traffic already diverted. That is why a housing has been radiographed and is on the deck.

What the chart says now. 82.9 kilometres along the route, at the repeater rather than between repeaters, with a tolerance of 400 metres from the slack and 200 from everything else. A position and an object, rather than a kilometre and a hope.

TAKES AS READ

a budget that closes leaves nothing that needs another cause

VERDICT 96W

The pump accounts for 3.6 decibels of the 4.1, leaving 0.5. That residue is inside what an amplifier does when its input changes and it gives a little less gain, so it needs no second cause. But saying so is a judgement rather than an arithmetic result, and it is recorded as such. The budget is now 21.3 decibels against an allowance of 24, which is 2.7 of margin. That is better than before the switch, and worse than the span had when it was new. The fresh pump is the last one in that housing.

TAKEAWAY 21W

A budget closes when every decibel is attributed, and what is left over is either a second cause or a measurement.

12.2 A hundred points before the ship

AMPLIFIER AND POWER HALL · A PERSON · SCIENCETANK

SCENE 41W

Tanaka is in the ship's store with a hundred points of spares and survey money left in the year, and four things it could go on. The ship sails in the morning and the next charter is not until the spring.

GUIDE 75W

What the station holds already. A pump output figure for repeater 6 that is four years old and was never re-measured. A slack figure that is an average over a whole route. An instrument whose index setting was wrong for eleven years and has now been corrected. And eleven mechanical joints on the shore end that are as old as the cable. The traffic is on a diverse route with no spare of its own.

ASK 7W

Spread a hundred points across the four.

TAKES AS READ

money spent on a measurement is worth what the measurement would change

VERDICT 100W

The optical telemetry is the measurement whose absence cost this fortnight. A pump current told everybody nothing for three weeks. The housing going to sea tomorrow is the only chance to add it. The slack survey turns the largest and least certain of the three corrections into a measurement where the ship will actually be working. Replacing the old joints is real work on eleven components that are past their design life and are not currently failing. A second instrument would halve the position uncertainty of a fault whose position is now dominated by slack rather than by the trace.

TAKEAWAY 19W

Evidence is worth what it would change about the next decision, not what it would add to the file.

12.3 One nucleus, one line

RADIOGRAPHY BAY · A ROOM · CASEBOOK

SCENE 30W

Farouk's source emits at two definite energies and nothing in between, however old it gets and however warm the bay is. Her film's exposure changes and the energies do not.

GUIDE 40W

Each observation on the left has one explanation on the right. What separates them is which properties belong to the nucleus itself, and which to how much of it is left. Ageing changes one of those and not the other.

ASK 10W

Match each observation about the source to what explains it.

TAKES AS READ

a nucleus has a set of states with definite energies · the nucleus, and the binding energy that holds it — taken as read

VERDICT 105W

Four observations, one distinction. The energies come from the gaps between the nucleus's own states, and nothing available in a radiography bay can alter those. So they are the same after eight months and the same when warm. The rate depends on how many nuclei are left, which is what has changed. The half-life belongs to that kind of nucleus and not to the pot, which is why it can be looked up rather than measured. And heat supplies fractions of an electron volt against nuclear energies of hundreds of thousands. That is why a nuclear source is the most stable thing on this site.

TAKEAWAY 18W

A nucleus has fixed levels, so it emits fixed energies; how much is left changes only how often.

END OF THE DAY 21W

A budget that closes against one cause and not the others is the strongest statement a fortnight of measurement can make.

THE CLOSING CARD

The ship sailed to 82.9 kilometres along the route rather than 84.6 along the fibre, and it went for a repeater rather than a break. Both halves of that were the fortnight: the loss was equal at both wavelengths, which no bend or crush can be, and switching repeater 6 to its spare pump returned 3.6 of the 4.1 decibels the span had lost. The position was 1.7 kilometres out for three reasons that all pointed the same way — the pulse travels at the group index and not the core index, there is more fibre than cable, and there is more cable than route. The housing on the deck was radiographed with a source at a tenth of its original strength, on an exposure of a hundred and twenty-six minutes, behind a barrier at eighteen and a half metres.

What it cost: the only spare pump in that housing, six days of charter, and a set of eleven-year-old joints that are still in the shore end. What is unfinished: the new housing went to sea with its pump current telemetered and its optical output not, which is the measurement whose absence cost three weeks; the slack is still an average over a whole route rather than a survey of the twenty kilometres the ship worked in; and nobody has measured this rope of glass — the stiffness of the argument, the index a pulse really travels at on this cable, is still the maker's figure.

You derived the acceptance cone, and then the distance, and then the three corrections that stood between a distance and a place. You found the ceiling on what a pump can ever give and used it to state a decline nobody could measure from shore. And you were the one who put the two-wavelength comparison beside the trace, which is what said the fibre was never damaged at all. A ship went to the right kilometre for the right object. That was your fortnight.

Card text only — no options, boards or answers. 8,574 words of card prose across 40 stops. Generated from themes/darkfibre by tools/make-cardbook.mjs.